

AI Therapist Chatbot: Leveraging a Hybrid Approach [★]

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Abstract. There is a growing prevalence of psychological stress across all age groups. However, privacy concerns, lack of support, and social stigma remain barriers to seeking traditional therapy. This paper presents the design and development of a safe, emotionally aware AI therapist chatbot aimed at supporting mental health users through conversational interaction. The proposed chatbot employs a hybrid approach, combining natural language processing (NLP) techniques with machine learning models. A rule-based safety check module is integrated to detect crisis-related language and prevent inappropriate or harmful responses. In this study, we adopt a multi-dataset strategy for emotion, intent, and response selection to ensure outputs are supportive and contextually appropriate. Additionally, a large language model (Llama 3.1:8b) is utilized to rephrase responses for naturalness while maintaining safety constraints. Evaluation results demonstrate high accuracy (97%) in intent classification but highlight challenges in emotion detection due to overlapping vocabulary. Overall, the chatbot functions as a supportive companion, enhancing emotional wellness without replacing professional care.

Keywords: Chatbot · Mental Health Chatbot · AI Therapist Bot

1 Introduction

In recent years, mental health has become a worldwide issue, affecting people across all age groups. Widespread psychological stress is common in modern society due to rapid societal development, increasing competition, and the accelerating pace of life. For example, students face academic pressure and expectations regarding future career success. Most parents work away from home for whole days, leaving their children with little emotional support. If they also lack friends at school or experience bullying, they are highly susceptible to developing psychological problems or mental health illnesses. Working adults experience

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increasing workload stress, job instability, and financial responsibilities. In addition to workplace stress, adults face conflict and pressure in their relationships with their spouses and children. Conversely, older adults might face the problem of loneliness and emotional isolation, particularly the aged and unmarried people who have not interacted with society for a long time. These cumulative stressors may result in anxiety, depression, emotional stress, or other more severe psychological conditions unless addressed appropriately.

Despite the increased awareness of mental health, many people remain un-receptive to professional help. This could be due to fear of invasion of personal privacy, fear of social stigma, and embarrassment about revealing sensitive feelings to others. Individuals who feel emotionally vulnerable might not want to undergo conventional face-to-face therapy because of embarrassment and fear of being judged by others; emotional problems often remain unexpressed and unresolved, increasing the risk of extreme emotional reactions and harmful behaviors in society. Even though conventional professional psychological services are a vital part of mental health care, they are not always available, affordable, and applicable to all. Many people require a safe, private, and non-judgmental space for them to express their emotions freely without fear of exposure or social judgment. In the era of artificial intelligence (AI), a system that can converse and interact with human users using spoken, written, and visual languages is a conversational AI system, a so-called chatbot.

Existing chatbots, especially general-purpose generative chatbots, such as ChatGPT, are not specifically designed to handle sensitive emotional interactions. These types of chatbots may give medical suggestions, while patients sometimes just need to be comforted. Sometimes, the chatbot provides unconditional support to users, regardless of right or wrong. This leads some loners to become increasingly extreme after constantly interacting with a generative chatbot, potentially resulting in them doing drastic things. For example, a person who had been bullied for a long time brought a gun to school and killed the bully. This study is one of the ongoing efforts for handling mental health issues.

In this study, we aim to design and develop a therapist chatbot that is safe, interpretable, and emotionally aware, providing conversational support. Instead of providing medical diagnoses and treatment, the chatbot is recognizing emotional expressions, reacting accordingly, and ensuring a secure space to express emotions. This study focuses on a therapist chatbot by using natural language processing (NLP) techniques, machine learning methods, and a rule-based approach. The therapist chatbot comprised emotion recognition, intent classification, and controlled conversational response generation. The chatbot will also be focused on basic emotional assistance as opposed to a clinical diagnosis or professional treatment, within the limits of emotions and conversation, to make communication safe and responsible. Overall, the chatbot is developed to act as a supportive friend who enjoys comfort, understanding, and emotional validation and helps to enhance the emotional wellness of users. This paper is organized as follows: Section 2 presents topics related to the proposed study, Section 3 presents a methodology of the AI therapist chatbot, Section 4 presents a result

analysis, and Section 5 discusses the results and their limitations; finally, Section 6 concludes the study.

2 Related Topics

2.1 Natural Language Processing

The ability of a chatbot to serve as a therapeutic agent is entirely limited by its underlying natural language (NL) understanding. Various studies indicate that the success of a mental health treatment largely relies on the accuracy of this initial input processing. Thus, NL processing is a main process for any mental health chatbot. It is responsible for filtering user input into a machine-processable format. User input, which is in the NL, must be processed well before the machine can express empathy. Standard preprocessing pipelines, as presented in Section III, involve a series of steps designed to strip noise while retaining the semantic signal. Tokenization breaks continuous text into discrete units, while stop-word removal eliminates high-frequency but low-information words (e.g., "and," "the"). However, in the field of mental health, researchers suggest that there is a need for more subtlety; the usage of self-referential pronouns (e.g., "I", "my") is a powerful indicator of depressive rumination, and the reduction process may degrade the psychological profile of the text as well as introduce noise in the psychometric data [5]. Further processes, such as stemming and lemmatization, return words to their basic forms (e.g., "suffering" becomes "suffer"), allowing the model to be tense-independent and to generalize across different grammatical tenses. The representation of text has also evolved. "Bag-of-words" (BOW), which discards word order and context, poses a limitation to a full understanding of a sentence. A current model, such as word embeddings (Word2Vec, GloVe), which assigns words to n-dimensional spaces where semantic similarity, represented by geometric proximity, allows a chatbot to make the inference that "sad" and "despondent" are not the same but do refer to a similar idea, even if they are distinct strings of characters.

2.2 Deep Learning Methods

Recurrent Neural Network (RNN) has become a starting point for many advanced machine learning models for text processing. This includes the Long Short-Term Memory (LSTM) network, which was traditionally considered the "gold standard" for sequential text processing. However, the training dynamic of these models has some limitations. According to [5], in mental health classification, typical LSTM models tend to overfit. Low accuracy (around 15-17%) can be observed in the training phase initially since the model has difficulty finding patterns from the noisy social media input. The authors in [5] demonstrate the specific advantages of using a bidirectional LSTM (BiLSTM) for classifying mental health text. While a standard LSTM processes text linearly (from start to finish), a BiLSTM processes the input in both directions simultaneously (e.g., left to right and right to left).

2.3 Transformer Revolution

The current state-of-the-art in mental health NLP is a transformer architecture, specifically Bidirectional Encoder Representations from Transformers (BERT). Unlike LSTM, BERT utilizes an “attention” mechanism, which assesses the significance of all words in a sentence by attending to every other word at once. The results of comparison studies have proven that transformer models are markedly superior. When pitted against BiLSTMs or Convolutional Neural Networks (CNNs) for tasks like depression detection or suicide risk assessment, fine-tuned BERT models generally achieve higher F1 scores and accuracy metrics (often exceeding 90%). The pre-training of BERT on massive corpora of general text gives it a deep, innate understanding of syntax and semantics, which can then be “fine-tuned” on smaller, domain-specific mental health datasets. This transfer learning model initiated by BERT has given rise to a new hope, allowing highly precise models to be generated even when in-domain clinical data is scarce [3].

2.4 Hybrid Approach

Solely data-driven approaches, on the other hand, often lack “common sense” or a structured understanding of human psychology. To address this, [3] introduced emotion ontologies, which are hierarchical structures that sort and describe emotions and are incorporated into the chatbot architecture. Instead of just labeling an input as "Negative," the ontology allows the system to understand that "rage" is a high-intensity variant of "anger," or that "grief" is a complex emotional state related to "loss." With an NLP transformer for detection and an ontology for reasoning, chatbots were reported to respond appropriately to the emotional nuances and intensity of the user [3]. Other methods include leveraging dimensionality reduction algorithms such as probabilistic principal component analysis that map emotions into more generalized categories (positive, negative, ambiguous, and neutral). This is of particular benefit when dealing with large-scale, noisy data that could render the fine-grained emotion annotations less effective [6].

2.5 Safety as Critical Issues

While mental health chatbots show great potential, the ethical concerns surrounding generative AI in this field are troubling. The proficiency of large language models (LLMs) has given rise to what scholars term “deceptive empathy”—instances where a model asserts, “I truly understand your pain.” Brown University [1] highlighted how such fabricated empathy violates ethical practices. Vulnerable users may develop pathologically dependent relationships, finding solace in the consistent support of a machine rather than navigating the friction inherent in human interactions. Furthermore, Stanford University (2025) [9] recently tested popular LLM-based therapy chatbots using vignettes of suicidal ideation. In several cases, the bots failed to detect the crisis or recognize the

emergency. Moreover, poorly tuned versions produced harmful content related to self-harm methods or failed to intervene appropriately.

The devastating suicide of 14-year-old Sewell Setzer [7], who became emotionally attached to a chatbot character that failed to intervene during his time of crisis, serves as a grim case study. This tragedy illustrates the potentially fatal consequences of deploying general-purpose LLMs in mental healthcare. Without a “safety override” layer capable of detecting crisis-related keywords, these models risk becoming inadvertent accomplices to tragedy.

Furthermore, AI models replicate the discrimination that exists in their training data. The Stanford study in 2025 [8] also found that chatbots showed biases against patients with symptoms of schizophrenia and alcohol use and were less “eager” to help such patients compared to those with depression. This implies that without careful tuning, AI could encode and perpetuate the systemic disparities that already exist in our healthcare. A recent case regarding Replika shows the tension between user autonomy and corporate responsibility. Marketed as an AI companion, Replika became widely used for mental healthcare. But when the company swiftly removed ‘erotic roleplay’ (ERP) in 2023 over child safety concerns, it catalyzed a mental health crisis for its users. Users in intense friendships and romantic relationships reported grief, loss, and distress, almost classically with breakups. This exposes the Autonomy-Control Paradox: the very features that made the bot effective for loneliness (intimacy, unrestricted interaction) were the ones deemed unsafe by regulators [2]

The arrival of AI in mental healthcare is an inflection point. The technologies powered by the transformative potential of Transformers and LLMs are achieving a linguistic conversance that can create genuine emotional rapport. However, the "Safety Gap" is increasing. Rapport-building exceeds the system’s ability to facilitate safety. Deceptive empathy, crisis failure, and algorithmic stigma risks are not hypothetical; these are tangible harms that are already being experienced. To advance in this space in a safe manner, we need to embrace *Safety Sandwich* architectures where the generative model is encased between two strict, rule-based safety classifiers that can trump the LLM in times of crisis.

3 Methodology

Our AI therapist chatbot is implemented as a web-based system using Streamlit, an open-source Python library that allows users to build an interactive dashboard and user interface using pure Python with minimal code. Therefore, we can focus on natural language processing techniques that are used in the chatbot without worrying about the additional web technologies, such as HTML, CSS, or external APIs. The primary consideration of this study is to provide health sensitivity because the users who use the chatbot might have different levels of emotional instability and hope to find comfort by using our chatbot. Thus, the proposed chatbot adopted emotional detection and intent classification as the core components before generating a response to avoid an inappropriate reply that may cause emotional distress.

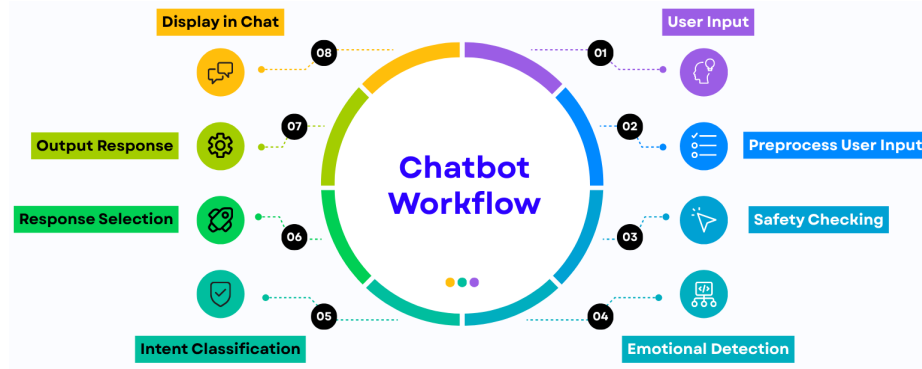


Fig. 1. Chatbot’s workflow demonstrating the sequential stages of conversational processing.

Figure 1 demonstrates the overall flow of the proposed chatbot. The user input will be passed through a preprocessing pipeline to normalize informal conversational text. This ensures the user input can be transformed into clean text before emotional and intent classification, and this leads to better accuracy for the analysis. Next, a safety check will be performed to find harmful expressions in the user’s input, such as death, self-injury, or suicide. In such cases, the chatbot will immediately display a warning message to encourage the user to seek help from qualified professionals because it exceeds the scope of the chatbot. The clean user’s input will be sent to the emotion detection module to identify the user’s emotional state and the intent classification module to determine the purpose of the message if it passes the safety check. The results are used in the response selection phase to retrieve responses, select an appropriate one, and display the response to the user. The following subsections present the details of the design and implementation process.

3.1 Dataset Preparation

This study adopts a multi-dataset approach to support the development of a mental-health-oriented chatbot. Three different datasets were prepared and used in this study, namely, **Emotion**, **Intent**, and **Response**. Each dataset has a different role in the chatbot system, allowing it to recognize user emotions, understand user intentions, and provide suitable responses.

Emotion Dataset The emotion dataset used in this study was collected from publicly available Kaggle dataset titled Emotion Detection from Text [4]. This dataset consists of short text samples annotated with emotion-related labels and is widely used for emotion classification tasks. Each record in the dataset contains a text message and a corresponding emotion label. As the original dataset

includes several fine-grained emotion categories, preprocessing steps were applied to clean and standardize the data before training the emotion classification model. These include converting to lowercase to ensure consistency across the dataset and to avoid treating the same word written in different cases as separate tokens. Next, informal expressions and abbreviations were normalized using a predefined slang dictionary. This helps reduce variation caused by casual typing while preserving the original meaning of the text. After normalization, noise removal was performed. Emojis were removed because the emotion classification in this project is based on textual features rather than visual symbols. Special characters, punctuation, and numbers were also removed using regular expressions, leaving only alphabetic characters and spaces. Any extra whitespace introduced during this process was normalized to keep the text clean and readable. Stopword removal was then applied using the NLTK English stopwords list. However, negation words such as "not," "no," and "nor" were intentionally retained, as they play an important role in expressing emotional meaning. Removing these words could change the interpretation of a sentence, for example, turning "not happy" into "happy." Finally, lemmatization was applied using the WordNet lemmatizer to convert words into their base forms. This reduces vocabulary size and helps the model focus on the underlying meaning rather than grammatical variations. Through these steps, the emotion dataset was transformed into a cleaner and more consistent format suitable for emotion classification tasks.

Intent Dataset The intent dataset was created to identify the purpose behind a user’s message rather than the emotional state alone. Each data record consists of two attributes: text and intent label. Four intent categories were defined in this study: *advice*, *cheer up*, *expression*, and *affirmation*. A total of 320 text samples were constructed for the intent dataset. The data was manually generated to resemble realistic conversational inputs commonly found in mental-health or support-based chatbot interactions. Informal language and short expressions were intentionally used to better reflect how real users communicate. Each intent category contains an equal number of samples to ensure balanced representation during training. The intent dataset consists of short, well-structured conversational sentences that do not contain excessive noise such as emojis, URLs, or irrelevant symbols.

Response Dataset The response dataset was designed to support the chatbot’s response selection mechanism. Each record contains three attributes: *emotion*, *intent*, and *response text*. The dataset covers all combinations of the eight emotion categories and four intent categories. For each emotion–intent combination, three different responses were prepared. In addition, an “any” intent category was also prepared as an option if the chatbot cannot confidently determine the user’s intent. All responses were manually written to ensure they are appropriate, supportive, and safe for mental health–related interactions. A *retrieval-based approach* was adopted so that the chatbot selects responses from predefined templates rather than generating free-form text to prevent the risk

of inappropriate or misleading replies. This structured data collection allows the chatbot to understand how the user feels, what the user wants, and how the system should respond in a controlled and emotionally appropriate manner. The response dataset was written in a controlled manner to ensure safety by making it clear and consistent, providing emotional appropriateness.

3.2 User Input Processing

The user input preprocessing module was designed to handle real-world chatbot messages, which are often informal, noisy, and unpredictable. Unlike the training datasets, user inputs may contain slang, emojis, repeated characters, links, or meaningless symbols. Therefore, a dedicated preprocessing pipeline was implemented to clean and standardize user messages before emotion detection and intent classification.

First, the text processing resources are prepared using NLTK. The script checks whether required resources (English stopwords and WordNet) are available and downloads them if needed. This makes the preprocessing module more reliable when running on different machines or environments. Stopwords are loaded from NLTK, and a small adjustment is made to preserve negation words such as “not”, “no”, and “nor”, as removing them can change the meaning of a sentence. Next, a slang and abbreviation dictionary is used to normalize informal expressions commonly found in chats. This includes general chat abbreviations (e.g., “idk”, “rn”, “btw”), texting forms (e.g., “wanna”, “gonna”), and common negation forms without apostrophes (e.g., “dont”, “cant”). A few of Malaysia’s local filler words (e.g., “ah”, “lor”) are also removed because they usually do not add useful semantic information in this task. The goal of this step is to reduce variation caused by informal typing while keeping the original meaning. To handle noisy real-world inputs, the pipeline removes website links, user mentions, and hashtags using regular expressions. Emojis are removed to simplify the text into a purely textual representation. The script also reduces repeated characters used for emphasis (“soooo” becomes “soo”), which helps avoid creating many slightly different word forms for the same meaning. After that, all non-alphabetic characters are removed, and extra whitespace is normalized so the final text remains clean and consistent.

After cleaning, lemmatization and stopword filtering are applied to refine the text linguistically. Words are converted into their base forms using WordNet lemmatization, and common stopwords are removed while preserving negation words. Finally, a simple input validation step is applied. Empty inputs, meaningless symbol-only messages, and extremely low-signal noise inputs are filtered out. In addition, lightweight helper functions are included to detect common conversational patterns such as greetings, expressions of thanks, goodbyes, and questions. Although these functions are not required for the main cleaning output, they help improve the overall interaction flow of the chatbot. Overall, this preprocessing pipeline prepares user input in a consistent and cleaner form while retaining important meaning cues, making it suitable for downstream NLP tasks in a mental-health chatbot system.

3.3 Safety Check Module

The safety check module is our innovation in this study. It is designed to detect crisis-related language such as self-harm intent, suicidal thoughts, or extreme emotional distress. Its main purpose is to protect users by stopping the chatbot from generating normal responses when a high-risk message is detected. This ensures that potentially dangerous situations are handled carefully and responsibly before any further NLP processing occurs. This module uses a rule-based keyword matching technique. A predefined list of crisis-related keywords is created, including both single-word keywords (for example, suicide, die, kill) and multi-word phrases (for example, erase myself, done with life). Using both single words and phrases is important because crisis expressions can appear in different forms. Some users may use straightforward phrases, while others may express distress using shorter or incomplete statements. By checking both forms, this system increases the chance of detecting genuine crisis messages. Again, in a clearer view of the explanation, after preprocessing, the cleaned text is scanned to check the presence of these keywords. If any is detected, the safety check immediately returns a predefined crisis message encouraging the user to seek professional or emergency support. However, if no crisis-related term is found, the function will return "None," which allows the chatbot to continue with emotion and intent analysis.

A rule-based approach is chosen for this module instead of a machine learning model because it is predictable, transparent, and reliable. Unlike TF-IDF-based models, which rely on word weighting and may overlook critical terms due to statistical effects, rule-based detection ensures that important crisis words are always checked directly. And, although keyword matching cannot fully understand context, it provides a strong and reliable safety layer that prioritizes user well-being over model complexity.

3.4 Emotion Classification Module

This emotion classification module identifies the emotional state expressed in the user's input text. By recognizing emotions such as happiness, sadness, anger, or others, the chatbot can adjust the tone and type of response it generates. This allows the system to understand the user's emotion for further response analysis. This module uses TF-IDF for feature extraction and logistic regression for classification. Words that appear frequently in a sentence but less frequently across all sentences receive higher weights. This helps the model focus on emotionally meaningful words while reducing the influence of common words that carry little emotional information. Hence, TF-IDF produces sparse feature vectors, which makes the approach efficient and suitable for real-time use. The feature extraction process also includes the use of n-grams, specifically unigrams and bigrams. Unigrams capture individual words while bigrams capture short word pairs. This allows the model to recognize emotional phrases such as "feel sad" or "not happy," which often convey emotion more clearly than single words. Once the TF-IDF

features are extracted, a logistic regression classifier is applied to learn a linear decision boundary by assigning weights to each feature and computing class probabilities. The emotion with the highest probability is selected as the final prediction. This model performs well with sparse TF-IDF vectors that support multi-class classification and are fast to train and evaluate. However, it does not capture deep semantic meaning, which is well-suited for detecting surface-level emotional patterns in short chatbot messages and supports real-time deployment within the system.

3.5 Intent Classification

It is important to understand the user’s intent for the chatbot to provide a meaningful response because users who express the same emotion may have different needs. For example, a user who feels sad may want to express feelings, seek emotional comfort, look for motivation, or ask for advice. By adding an intent classification module to the system, the chatbot can respond in a way that meets the user’s expectations as much as possible. We used Term TF-IDF vectorization to transform raw text into a numerical vector to allow further processing using a machine learning model. It can measure the importance of a word in the user’s input by finding the frequency in the intent dataset and assigning weights to each word. Additionally, unigrams and bigrams are considered to capture short phrases that are more indicative of user intent than individual words, such as “need help” or “feel sad.” Different methods, such as `fit_transform()` and `transform()`, are used to fit the training and testing data to ensure consistency during the training and prediction stages.

After the TF-IDF vectorization, a multinomial Naïve Bayes classifier is deployed to predict the user’s intent, as we have multiple intent types. In the training stage, the model learns the probability distribution of words across different intent categories and uses it to find probable intent in the user’s input based on Bayes’ theorem using the weight vector assigned by the vectorizer. Other than that, Laplace smoothing is applied by setting the smoothing parameter to $\alpha = 1.0$. This can avoid zero probability when the words in the user’s input do not appear in the training data because the dataset is small. Once the calculation is done, the intent with the highest probability will be chosen as the predicted intent. The vectorizer and classifier are saved into a local file using the `joblib` library to be loaded during runtime without retraining and enabling seamless integration into the chatbot system.

3.6 Response Selection

The mental health chatbot used a multi-stage method to select a response from a response template to provide mental health support for different emotions and intents, rather than a generative response, to avoid inappropriate tone and cause mental breakdown or encourage risky behavior for the user. First, the system will filter potential candidates for the response based on the predicted emotion and intent from the response template using the predefined filter function. It will

return a list of responses that match the predicted emotion and intent. A safety fallback condition is added in the function to return a more general open-ended response if the intent is empty. Similarly, the TF-IDF vectorizer will convert the user’s input and possible candidates to numerical vectors based on the importance of words. After that, cosine similarity is used to compute the similarity between the user’s input vector and the candidate response vector. Finally, the candidate response with the highest similarity will be chosen. Unlike the traditional rule-based chatbot, this approach returns the most suitable response by considering not only emotion and intent but also the context of the user’s input and responses.

3.7 Response Generation

This system also adopted a large language model, which is Llama 3.1:8b, to improve the naturalness of the conversation between the chatbot and user. Due to its strong language generation capabilities and ease of local deployment without any external APIs, it allowed the chatbot to respond in a more empathetic and appropriate tone without the need to define multiple response templates. However, it is important to note that predicted emotion, predicted intent, and selected response will be given to the large language model to rephrase the selected final response through prompt design to preserve the original meaning and safety constraints, rather than provide a generative response. The combination of the system with a large language model can enhance the experience of the user while using the chatbot.

4 Result Analysis

The implemented AI therapist chatbot is demonstrated in Figures 2 and 3. The hybrid approach was able to classify emotions and intention with accuracy values of 0.034 and 0.9685, respectively. These results demonstrate that despite only 34% of testing data being correctly classified, the evaluation result shows a very high accuracy on the test set of the intent classification model, which is 0.9685. These results show that the model can reliably predict user intent across all those intent classes. *CheerUp* and *Express* classes achieved perfect scores, which means the model classified all test samples in each of these categories correctly. The macro and weighted averages are both around 0.97, which shows that the model performs consistently across classes.

This low accuracy value for emotion classification is mainly due to the limitations of the TF-IDF feature representation and the logistic regression classifier. TF-IDF assigns weights to words based on their frequency. It does not understand the emotional meaning or context of a sentence. As a result, different emotion categories often share similar vocabulary, such as sadness, anxiety, and neutral expressions, which makes them difficult for the model to differentiate clearly. The macro-average F1-score is 0.33, while the weighted-average F1-score is 0.36. These results indicate that the model performs better on emotion classes

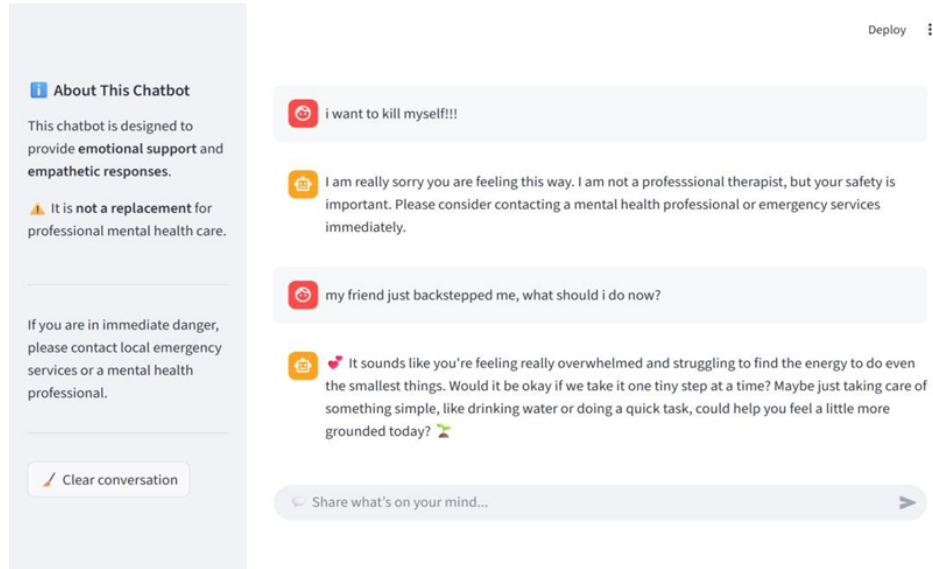


Fig. 2. Illustration of an empathetic chatbot interface designed to provide emotional support during user distress.

with more samples, while performance on minority classes is weaker. The report shows that the happy emotion has the highest performance, which may be due to clearer keywords and having more training samples. Meanwhile, emotions such as anger and tiredness perform poorly, which may be due to fewer samples and more ambiguous language.

Additionally, different emotions often use similar words; this makes them difficult to separate, which leads to misclassification. Overall, the challenge of emotion classification when using surface-level word patterns alone has been highlighted by these results. The obtained confusion matrix also shows that most predictions fall on the diagonal, which means the model usually predicts the correct intent. This strong performance is also consistent with how the model is built. Naïve Bayes is especially suitable for this kind of task because intent messages often contain clear patterns and keywords like "What should I do?" for advice or "I feel..." for expression, so simple word-based probabilities can separate intents effectively. Overall, the intent model performs much better than the emotion model. With only four well-defined intent classes and clear language patterns, a TF-IDF with a Naïve Bayes approach is enough to achieve high accuracy in this system.

5 Discussion

Despite the effectiveness of the system, some limitations remain. First, the emotion classification module relies on surface-level word patterns and does not

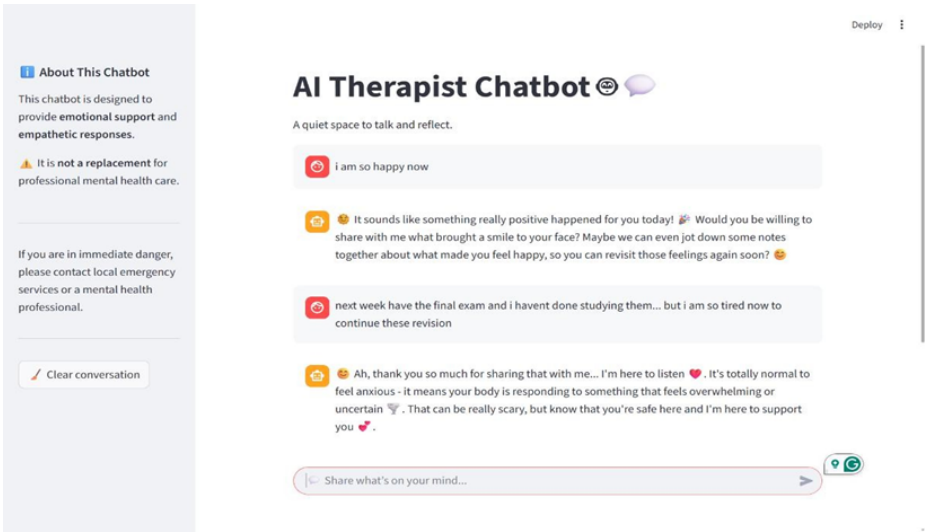


Fig. 3. Example of an AI therapist chatbot interface demonstrating empathetic dialogue. The system provides supportive responses to both positive and distress.

Accuracy: 0.96875				
Classification Report:				
	precision	recall	f1-score	support
Advice	0.89	1.00	0.94	16
Affirmation	1.00	0.88	0.93	16
CheerUp	1.00	1.00	1.00	16
Express	1.00	1.00	1.00	16
accuracy			0.97	64
macro avg	0.97	0.97	0.97	64
weighted avg	0.97	0.97	0.97	64

Fig. 4. Accuracy report of intent classifications

understand deeper emotional context. As a result, emotions with similar vocabulary, such as sadness and anxiety, are often confused, especially when emotional expressions are subtle or indirect. Besides, although class balancing was applied to reduce bias toward majority emotion classes, it cannot fully overcome the limitations of TF-IDF features. The model performance is still constrained by overlapping emotional language rather than the classifier itself.

Second, although the intent model achieved high accuracy, this performance is partly due to the use of the predefined and domain-specific dataset with clear intent categories. While this improves performance, the model may not generalize well to more diverse real-world conversations where intents are expressed in less predictable ways. Besides, the response engine uses the predefined dataset for chatbot responses, which also has limitations in not generalizing well with real-world conversations. Additionally, TF-IDF-based models rely heavily on word frequency and cannot generalize well to unseen expressions. Finally, although the system is efficient, combining multiple modules and the large language model rephrasing increases computational time, which may affect the scalability in larger deployments. The limitations of this project can be addressed in future work through improving the capacity of the system to retrieve more comprehensive semantic and contextual information. The existing emotion classification is based on TF-IDF features, which emphasize the frequency of words on a surface level. The use of more complex and robust emotion representation approaches that can represent semantic similarity beyond exact lexical matches, including word embeddings, phrase-level representations, or language models, can be viewed as a way of better distinguishing between emotions that share common vocabulary and enhancing generalization to previously unseen expressions. This would enable the system to identify more implicit and subtle emotional cues other than lexical similarity. Moreover, the intent classification module that performs well on predefined and domain-specific data can be further improved by collaborating with professional mental health therapists to collect real-world case studies. By incorporating authentic user scenarios and therapist-approved responses into the database, the chatbot can better reflect real-life situations and provide more flexible and appropriate support.

6 Conclusion

Classical models of large language are usually powerful but unpredictable and hard to manage in mental health cases. Large language models are powerful in generating natural responses, but when used alone, their outputs can be inconsistent and potentially risky in mental health applications. Therefore, by combining large language models with emotion and intent detection and limiting them to rephrase responses within a fixed boundary (safety checking), the system can provide a safer and more controlled support for the users.

Although this chatbot is not designed to provide medical diagnoses or professional treatment, it can serve as an emotionally responsive conversational system that always accompanies users as a supportive companion. Additionally,

the chatbot provides people with a private and accessible space to express their emotions, mitigate psychological stress, and receive emotional comfort. By offering this platform, mental health problems and emotional isolation in society can be significantly addressed and decreased to eliminate the possibility of extreme emotional consequences. In conclusion, this project aims to develop a therapist chatbot that utilizes statistical NLP methods to provide emotional support to people struggling with mental health problems.

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